PSTAT 126 Final Project

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Part 1: Data Description and Descriptive Statistics

1. Select a Random Sample

```
df <- read.csv("Diamonds Prices2022.csv")

# Remove index column if it exists
if(names(df)[1] == "" | names(df)[1] == "X") {
  df <- df[,-1]
}

set.seed(123)
sample_df <- df[sample(nrow(df), 1500), ]
head(sample_df)
```

```
      carat   cut color clarity depth table price    x    y    z
51663  0.73 Ideal     I     VS1  60.7    56  2397 5.85 5.81 3.54
2986   0.70 Ideal     G     VS1  60.8    56  3300 5.73 5.80 3.51
29925  0.31 Ideal     D     VS1  61.6    55   713 4.30 4.33 2.66
29710  0.31 Ideal     H    VVS1  62.2    56   707 4.34 4.37 2.71
37529  0.31 Ideal     E      IF  60.9    55   987 4.39 4.41 2.68
2757   0.83  Good     E     SI1  63.7    59  3250 5.95 5.89 3.77
```

2. Describe All Variables

The data set contains the following variables:

`row_id`: observation index/identifier

`carat`: weight of the diamond

`cut`: quality of cut

`color`: diamond color grade

`clarity`: clarity grade

`depth`: total depth percentage

`table`: width of top of diamond relative to widest point

`price`: price in dollars

`x`: length in mm

`y`: width in mm

`z`: depth in mm

```
# Summary and Structure
summary(sample_df)
```

```
     carat             cut                color              clarity
 Min.   :0.2300   Length:1500        Length:1500        Length:1500
 1st Qu.:0.4000   Class :character   Class :character   Class :character
 Median :0.7100   Mode  :character   Mode  :character   Mode  :character
 Mean   :0.7963
 3rd Qu.:1.0400
 Max.   :2.4800
     depth           table           price               x
 Min.   :55.20   Min.   :52.00   Min.   :  364.0   Min.   :3.920
 1st Qu.:61.10   1st Qu.:56.00   1st Qu.:  959.5   1st Qu.:4.710
 Median :61.90   Median :57.00   Median : 2471.0   Median :5.710
 Mean   :61.83   Mean   :57.39   Mean   : 3924.4   Mean   :5.731
 3rd Qu.:62.50   3rd Qu.:59.00   3rd Qu.: 5292.0   3rd Qu.:6.530
 Max.   :79.00   Max.   :73.00   Max.   :18706.0   Max.   :8.640
       y               z
 Min.   :3.960   Min.   :0.000
 1st Qu.:4.720   1st Qu.:2.910
 Median :5.710   Median :3.535
 Mean   :5.732   Mean   :3.541
 3rd Qu.:6.530   3rd Qu.:4.032
 Max.   :8.580   Max.   :5.360
```

```
str(sample_df)
```

```
'data.frame':   1500 obs. of  10 variables:
 $ carat  : num  0.73 0.7 0.31 0.31 0.31 0.83 0.51 0.7 0.4 1.1 ...
 $ cut    : chr  "Ideal" "Ideal" "Ideal" "Ideal" ...
```

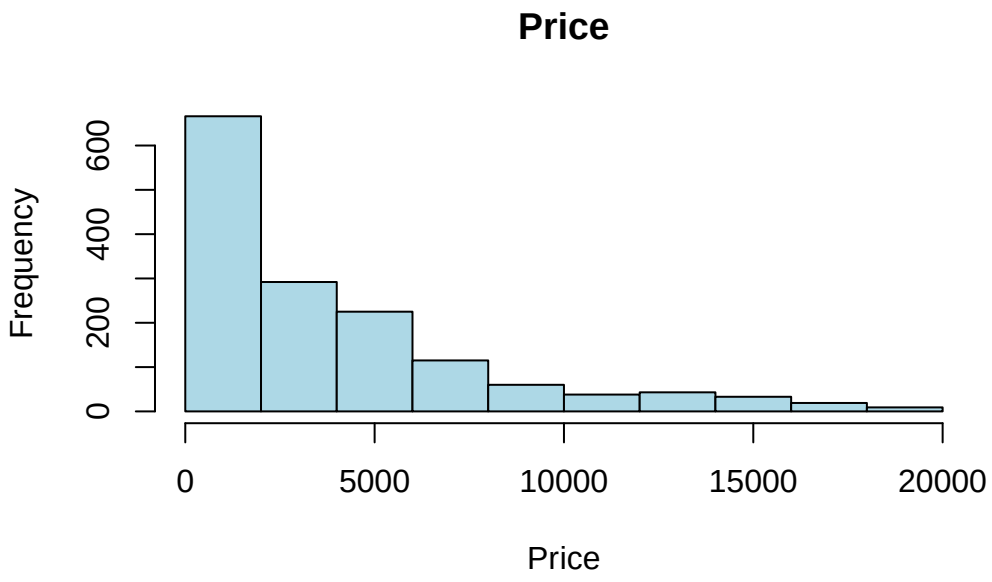
```
 $ color  : chr  "I" "G" "D" "H" ...
 $ clarity: chr  "VS1" "VS1" "VS1" "VVS1" ...
 $ depth  : num  60.7 60.8 61.6 62.2 60.9 63.7 62.5 64.2 61.6 61.2 ...
 $ table  : num  56 56 55 56 55 59 58 58 56 61 ...
 $ price  : int  2397 3300 713 707 987 3250 1668 1771 1053 4640 ...
 $ x      : num  5.85 5.73 4.3 4.34 4.39 5.95 5.12 5.59 4.73 6.61 ...
 $ y      : num  5.81 5.8 4.33 4.37 4.41 5.89 5.18 5.62 4.78 6.66 ...
 $ z      : num  3.54 3.51 2.66 2.71 2.68 3.77 3.22 3.6 2.93 4.01 ...
```

```r
# Reset graphics device and set smaller margins to prevent "margins too large" error
if(!is.null(dev.list())) dev.off()
```

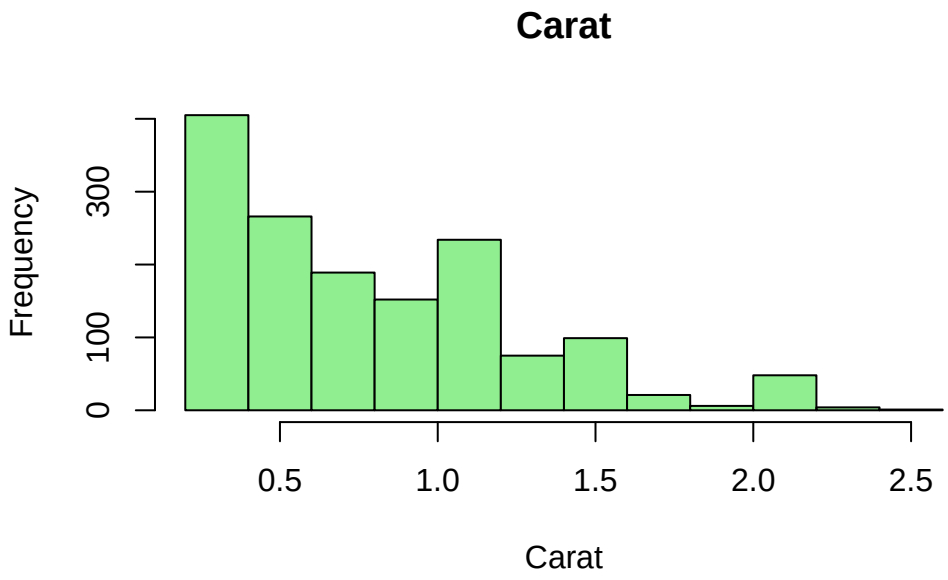
```
null device
          1
```

```r
par(mfrow=c(2,2), mar=c(4,4,2,1))
```

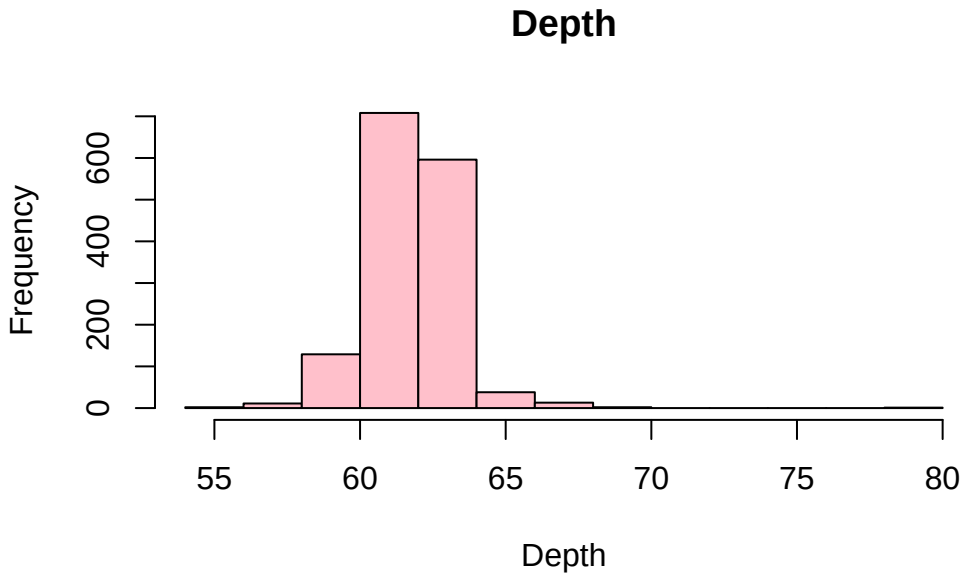
```r
# Histograms for continuous variables
hist(sample_df$price, main="Price", col="lightblue", xlab="Price")
```



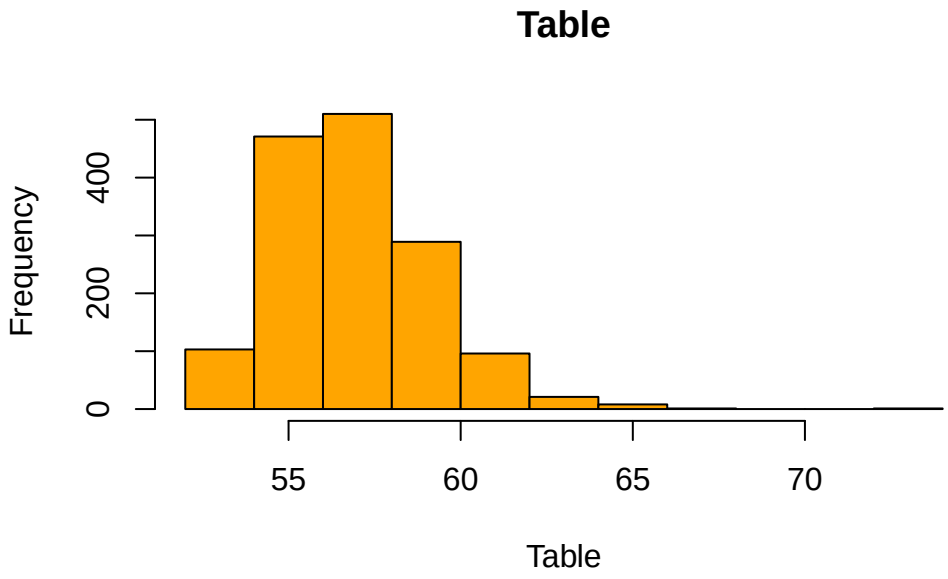
```r
hist(sample_df$carat, main="Carat", col="lightgreen", xlab="Carat")
```




```r
hist(sample_df$depth, main="Depth", col="pink", xlab="Depth")
```

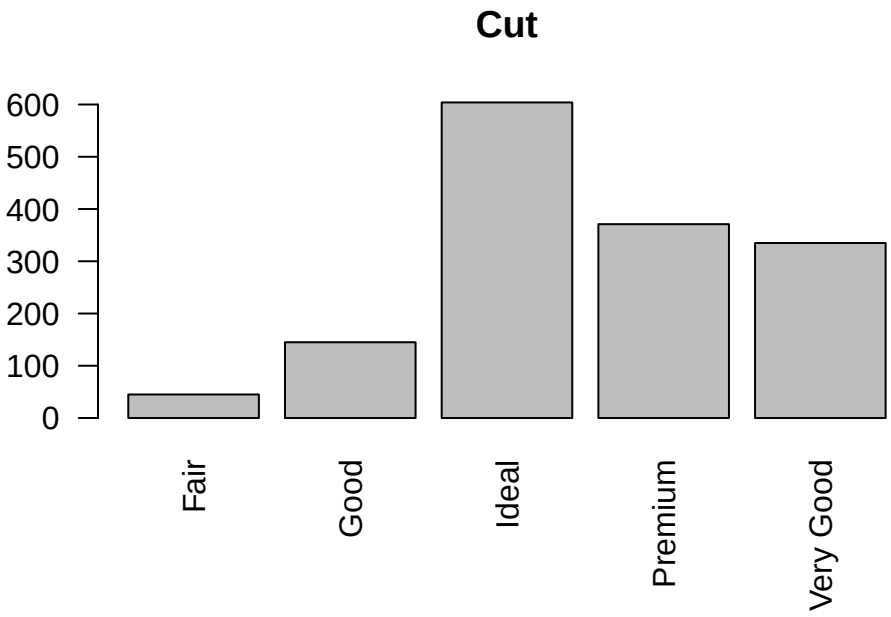


```r
hist(sample_df$table, main="Table", col="orange", xlab="Table")
```

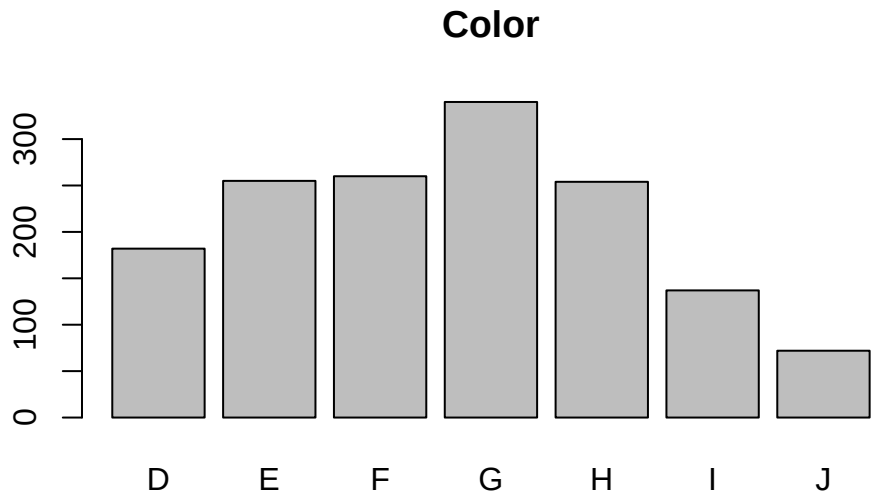



```r
# Bar plots for categorical variables
par(mfrow=c(1,3), mar=c(6,4,2,1))
```

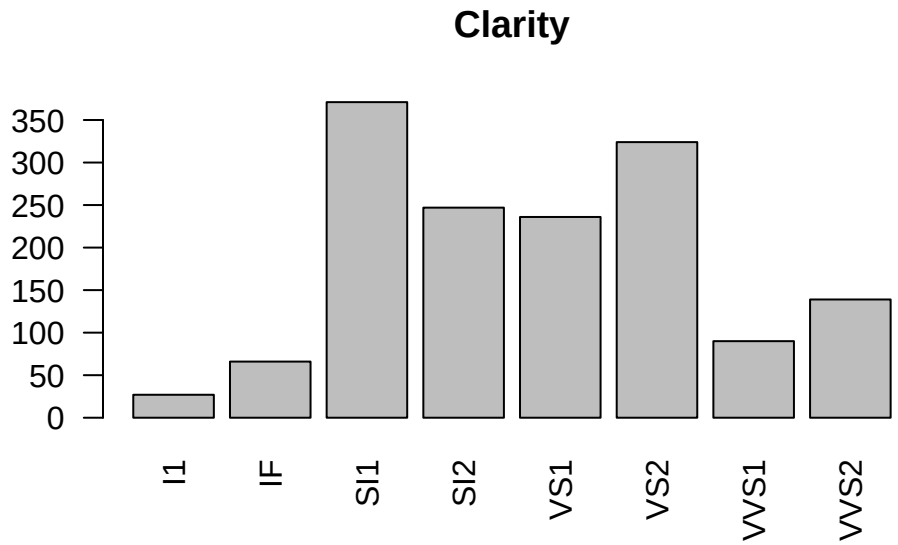
```r
barplot(table(sample_df$cut), main="Cut", las=2, col="grey")
```



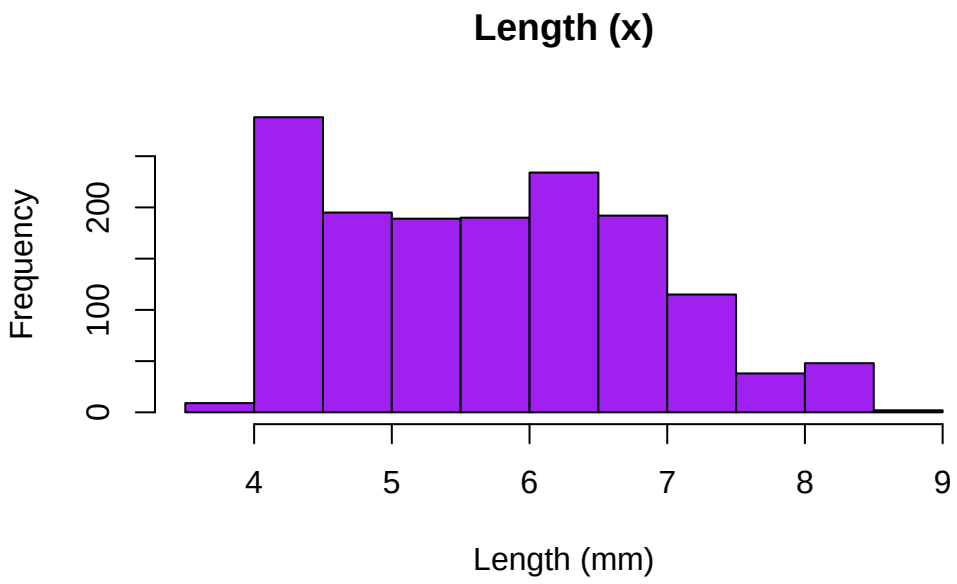

```r
barplot(table(sample_df$color), main="Color", col="grey")
```



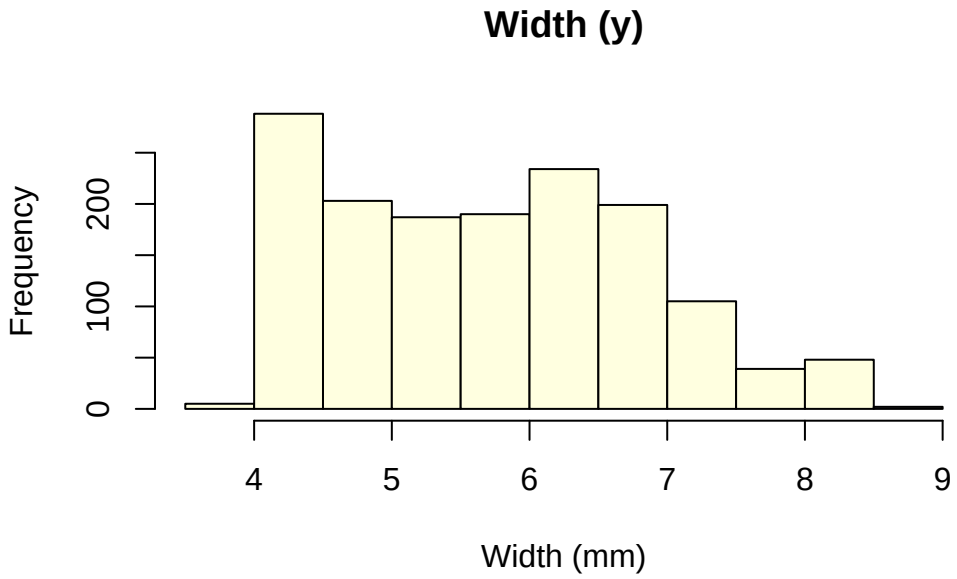
```r
barplot(table(sample_df$clarity), main="Clarity", las=2, col="grey")
```



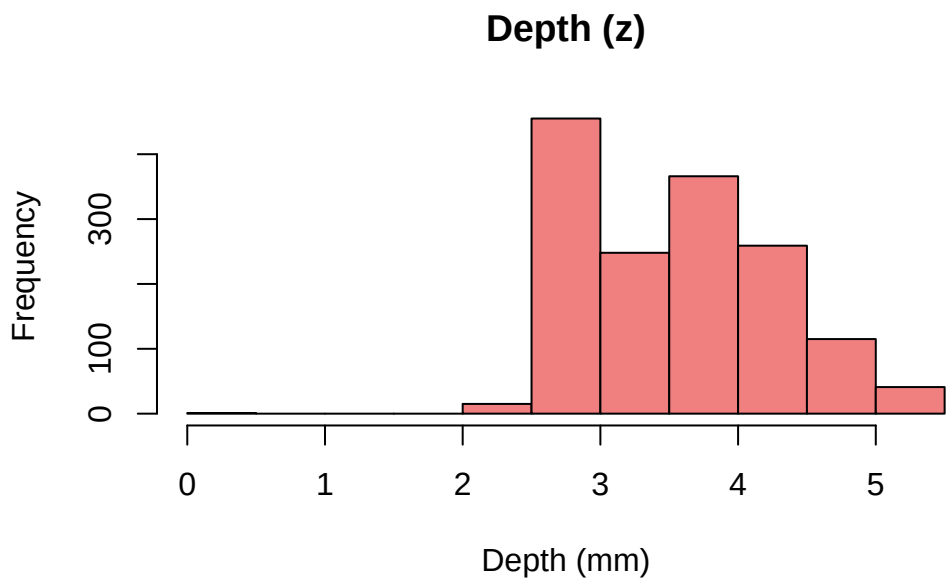

```r
# Histograms for x, y, z (diamond dimensions in mm)
hist(sample_df$x, main="Length (x)", col="purple", xlab="Length (mm)")
```




```r
hist(sample_df$y, main="Width (y)", col="lightyellow", xlab="Width (mm)")
```



```r
hist(sample_df$z, main="Depth (z)", col="lightcoral", xlab="Depth (mm)")
```



Comments: The price and carat variables are heavily right-skewed. Most diamonds in the sample are below 1 carat and under $5,000. Depth follows an approximately normal (bell-shaped) distribution, with the majority of values concentrated between 60 and 63. The table variable is slightly right-skewed, with most diamonds having a table value between 55 and 62 and a small number of extreme values above 65. The color distribution is roughly uniform across grades D through H, with fewer diamonds at the extreme ends (J being least common). The clarity distribution shows SI1 and VS2 as the most frequent grades, with I1 (lowest clarity) and IF (highest clarity) being the rarest.

3. Determine if there is any correlation between these variables.

```r
# Correlation for numeric variables only
numeric_cols <- sapply(sample_df, is.numeric)
cor(sample_df[, numeric_cols])
```

```
           carat         depth      table       price             x
carat 1.00000000  0.0468169639  0.1844685 0.920457651  0.9805666559
depth 0.04681696  1.0000000000 -0.1831374 0.009186303 -0.0003697883
table 0.18446849 -0.1831374045  1.0000000 0.126209814  0.2052152789
price 0.92045765  0.0091863034  0.1262098 1.000000000  0.8815469899
x     0.98056666 -0.0003697883  0.2052153 0.881546990  1.0000000000
```

```
y     0.98037087 -0.0043338848  0.2009565 0.884258147  0.9989320142
z     0.96907007  0.1325712817  0.1731332 0.870400357  0.9800049128
                y         z
carat  0.980370868 0.9690701
depth -0.004333885 0.1325713
table  0.200956453 0.1731332
price  0.884258147 0.8704004
x      0.998932014 0.9800049
y      1.000000000 0.9796798
z      0.979679752 1.0000000
```

There is definitely correlation between several numerical variables. From the sample, the strongest relationships were:

carat and price: strong positive correlation (about 0.92).

carat with x, y, and z: very strong positive correlations.

x and y: almost perfectly positively correlated.

depth: had only a weak relationship with price.

table: had a weak positive relationship with price.

This means diamond size variables move together strongly, and larger diamonds generally cost more.

4. Run the multiple linear regression model using all these variables and observe the summary statistics.

```r
full_model <- lm(price ~ ., data = sample_df)
summary(full_model)
```

```
Call:
lm(formula = price ~ ., data = sample_df)

Residuals:
    Min      1Q  Median      3Q     Max
-5896.4  -581.6  -171.2   416.2  6422.7

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   3095.48    2475.58   1.250 0.211350
carat        13285.10     332.38  39.969  < 2e-16 ***
```

```
cutGood        260.70     211.28   1.234 0.217430
cutIdeal       667.33     213.00   3.133 0.001764 **
cutPremium     445.09     203.22   2.190 0.028664 *
cutVery Good   504.53     209.45   2.409 0.016126 *
colorE        -247.12     107.22  -2.305 0.021311 *
colorF        -320.78     107.26  -2.991 0.002830 **
colorG        -527.40     103.02  -5.119 3.47e-07 ***
colorH       -1114.12     109.39 -10.184  < 2e-16 ***
colorI       -1688.70     128.28 -13.164  < 2e-16 ***
colorJ       -2444.93     157.30 -15.543  < 2e-16 ***
clarityIF     4161.14     269.59  15.435  < 2e-16 ***
claritySI1    2804.58     234.61  11.954  < 2e-16 ***
claritySI2    1869.32     236.79   7.894 5.64e-15 ***
clarityVS1    3812.35     239.73  15.903  < 2e-16 ***
clarityVS2    3616.82     236.72  15.279  < 2e-16 ***
clarityVVS1   4241.61     260.00  16.314  < 2e-16 ***
clarityVVS2   4180.08     248.26  16.838  < 2e-16 ***
depth          -26.89      29.68  -0.906 0.365022
table          -14.27      16.48  -0.866 0.386663
x            -2347.18     661.24  -3.550 0.000398 ***
y              936.92     661.31   1.417 0.156762
z             -630.87     286.90  -2.199 0.028038 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1095 on 1476 degrees of freedom
Multiple R-squared:  0.9249,    Adjusted R-squared:  0.9237
F-statistic: 790.3 on 23 and 1476 DF,  p-value: < 2.2e-16
```

5. Comment on anything of interest that occurred in this part. Were the data approximately what you expected, or did some of the results surprise you?

One interesting result is how strongly carat appears to be related to price. This is expected since larger diamonds usually cost more than smaller ones, but the relationship is still very strong in the sample. Another interesting point is that the variables x, y, and z are highly related to each other, which makes sense because they are all measurements of diamond dimensions. I also noticed that variables like depth and table don't appear to have as strong of a relation with price as carat does. Overall, the results were mostly what we all expected. The thing that stood out to us the most is the strength of the relationship between size-related variables.

Part 2

1. Start with one predictor and one response from the variables in Part 1. For instance, you can start with the predictor 'carat' and the response 'price', and

conduct a simple linear regression analysis on it.

```r
slr_model <- lm(price ~ carat, data = sample_df)
summary(slr_model)
```

```
Call:
lm(formula = price ~ carat, data = sample_df)

Residuals:
    Min      1Q  Median      3Q     Max
-5988.2  -874.0    -9.7   597.1  8675.4

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -2365.58      79.77  -29.65   <2e-16 ***
carat        7899.45      86.66   91.15   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1550 on 1498 degrees of freedom
Multiple R-squared:  0.8472,    Adjusted R-squared:  0.8471
F-statistic:  8308 on 1 and 1498 DF,  p-value: < 2.2e-16
```

The simple linear regression shows a strong positive relationship between carat and price, where each additional carat increases the predicted diamond price by about $7,899. The model explains approximately 84.7% of the variability in price ($R^2 = 0.847$).

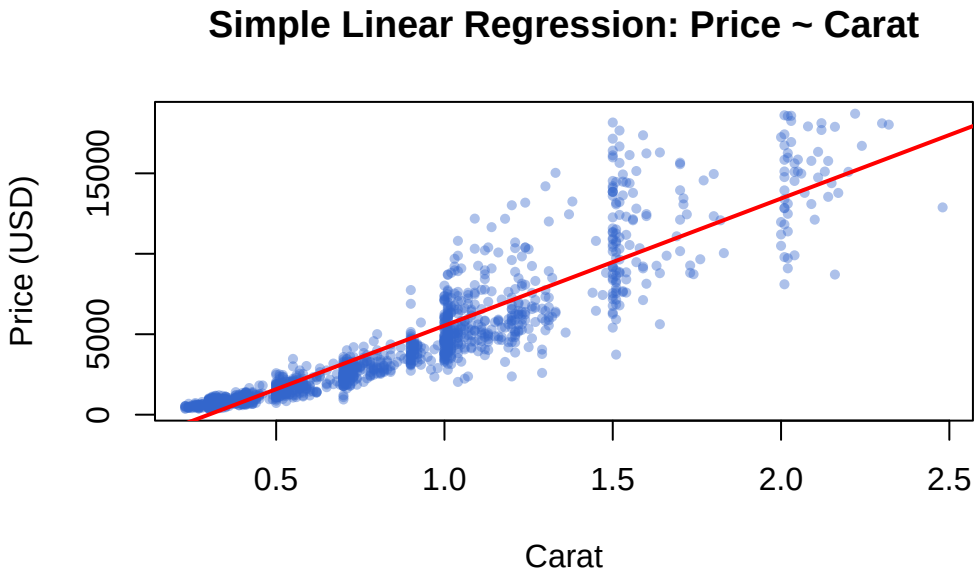
2. Run the model and examine the summary statistic, interpreting everything (hypothesis testing, R^2adj as discussed in class, confidence interval, predictor interval, plot, etc).

Interpretation of Simple Linear Regression Output: The model estimates that for every one-unit increase in carat, the price of a diamond increases by approximately $7,900 on average. The intercept of -2365 is not meaningful on its own since a diamond cannot have zero carats. Both coefficients are highly significant (p<2e-16), meaning we reject the null hypothesis that carat has no effect on price. The adjusted R^2 of 0.8471 tells us that carat alone explains about 85% of the random variation in diamond price, which is a strong result of a single predictor. The residual standard error of 1550 represents the average deviation of the actual prices from the predicted prices.

H0: Carat has no linear effect on price

H1: Carat has a significant linear effect on price

```r
#Scatterplot with regression line
plot(sample_df$carat, sample_df$price,
     xlab = "Carat", ylab = "Price (USD)",
     main = "Simple Linear Regression: Price ~ Carat",
     col = rgb(0.2, 0.4, 0.8, 0.4), pch = 16, cex = 0.7)
abline(slr_model, col = "red", lwd = 2)
```




```r
#Confidence Interval for the slopes
confint(slr_model)
```

```
                2.5 %    97.5 %
(Intercept) -2522.055 -2209.097
carat        7729.452  8069.443
```

```r
#Prediction Interval for a 1 carat diamond
new_data <- data.frame(carat = 1)
predict(slr_model, newdata = new_data, interval = "prediction", level = 0.95)
```

```
       fit      lwr      upr
1 5533.871 2492.097 8575.646
```

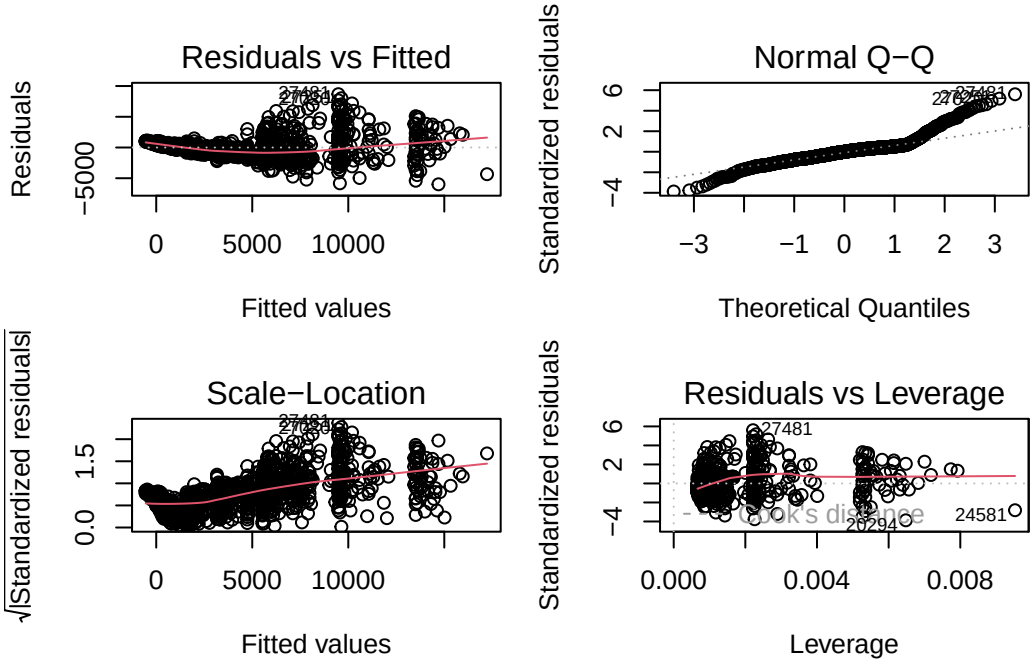
Interpretation: The scatterplot shows a clear positive relationship between carat and price- as carat increases, price increases. However, the spread of points widens as carat increases, which suggests non-constant variance (heteroscedasticity) and hints that a straight line may not be the perfect fit.

Confidence Interval: We are 95% confident that for each additional carat, the average price increases by between $7,729 and $8,069

Prediction Interval: For a 1 carat diamond specifically, the model predicts a price of 5,559, with a 95% prediction interval of $2,492 to $8,575. This wide range reflects high variability in individual diamond prices even at the same carat weight.

3. Test the assumptions and apply any necessary transformations to the response variable y or the predictor.

```
par(mfrow = c(2, 2), mar = c(4, 4, 3, 1))
plot(slr_model)
```



These plots confirm that all 3 major assumptions are violated. The Residual vs Fitted Plot shows us that the residuals fan outward as fitted values increase- this is heteroscedasticity (non constant variance). The red line also curved, indicating non-linearity. - The QQ Residuals plot shows that points deviate heavily from the diagonal at both tails (the normality assumption is violated). - The Scale-Location plot shows an upward trend in the red line confirming variance is not constant across fitted values - The Residuals vs Leverage point shows a few

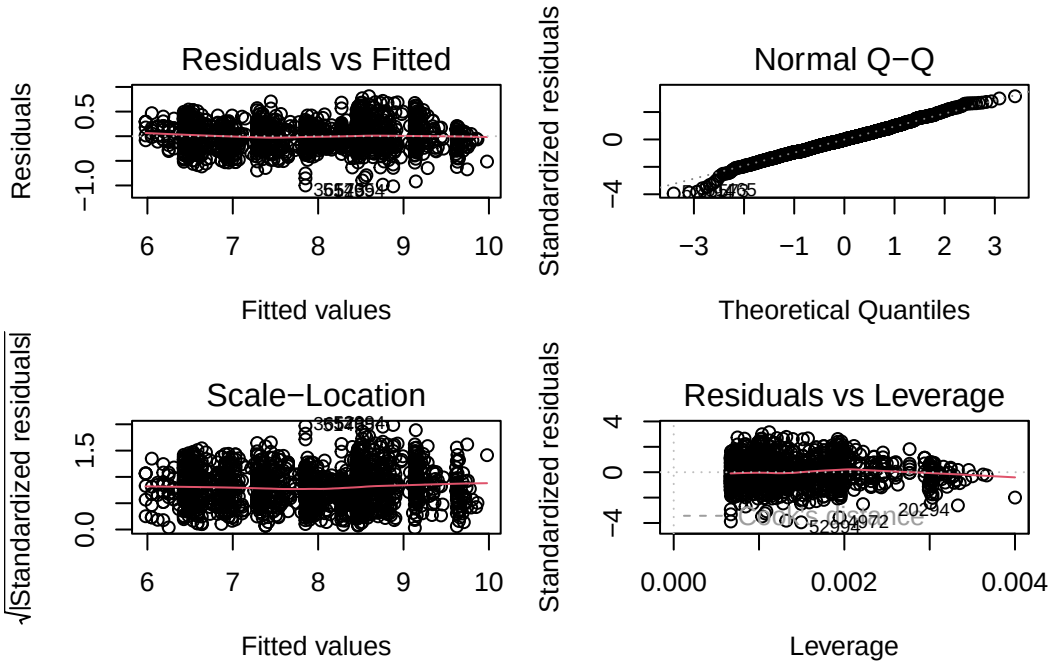
points (21567…) with high leverage but none cross Cook's distance, so no extreme outliers need to be removed.

Since all 3 assumptions fail, we run the transformation code

```r
#Applying Log Transformation to both variables
sample_df$log_price <- log(sample_df$price)
sample_df$log_carat <- log(sample_df$carat)

#Refitting model with transformed variables
log_model <- lm(log_price ~ log_carat, data = sample_df)

#Checking diagnostics again after transformation
par(mfrow = c(2, 2), mar = c(4, 4, 3, 1))
plot(log_model)
```



After applying the log transformation to both price and carat, all 3 previously violated assumptions are now sufficiently satisfied. The residuals scatter randomly with consistent spread, normality is greatly improved, and no influential outliers are present, making the transformed model valid for interpretation and inference.

4. Call the summary function on the transformed variables, observed summary, and note any changes

```r
summary(log_model)
```

```
Call:
lm(formula = log_price ~ log_carat, data = sample_df)

Residuals:
     Min       1Q   Median       3Q      Max
-1.01717 -0.16773 -0.00439  0.16296  0.81228

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  8.45080    0.00804    1051   <2e-16 ***
log_carat    1.68118    0.01144     147   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2579 on 1498 degrees of freedom
Multiple R-squared:  0.9352,    Adjusted R-squared:  0.9351
F-statistic: 2.162e+04 on 1 and 1498 DF,  p-value: < 2.2e-16
```

The log-transformed model shows meaningful improvements across every metric compared to the original non-transformed model. The adjusted R^2 jumped from 0.847 to 0.9352 meaning the transformed model now explains about 93.6% of the variation in log(price)- an 8.5 percentage point improvement just from transforming the variables. The slope changed to 1.681 meaning that for every 1% increase in carat weight, price increases 1.698%. This also implies that doubling the carat weight multiplies the price by roughly 3.2x, which reflects the real-world premium placed on larger diamonds. The intercept of 8.450 is now on the log scale. Back-transforming it gives 4,727, which represents the predicted price of a 1-carat diamond (since log(1) = 0). This is a much more interpretable and realistic baseline than the original intercept of -2370. The residual standard error dropped dramatically from 1,550 to 0.263 (on the log scale), confirming that individual predictions are now much closer to the fitted line on average. Both coefficients also remain highly statistically significant with p-values <2e-16, so carat remains a strong and reliable predictor of price even after the transformation.

5. Add other variables to the model and assess if the model improves. For step 5, run the code in the background and include all interpretations in the file. For instance, if adding depth to the simple linear regression model (carat and price) increases for adjusted R^2, include it in the model; if it decreases, exclude it. Do not include the code step 5 in the submitted file; only write the conclusions.

Starting from the transformed model log(price) ~ log(carat) with an adjusted R^2 of 0.9351, additional variables were added one at a time. Adding depth increased adjusted R^2 to 0.9360

and was kept. Adding table increased adjusted R^2 to 0.9367 and was kept. Adding cut increased adjusted R^2 to 0.9391 and was kept. Adding color increased adjusted R^2 to 0.9499 and was kept. Adding clarity increased adjusted R^2 to 0.9833 - the single largest improvement of any variable - and was kept. Adding x, y, and z together only increased adjusted R^2 by 0.0004, a negligible gain that does not justify adding three variables that introduce multicollinearity with log(carat), so they were excluded.

Final model: log(price) ~ log(carat) + depth + table + cut + color + clarity, with an adjusted R^2 of 0.9833, explaining approximately 98.3% of the variation in log(price).

6. Comment on anything of interest that occured while doing this part.

The most interesting finding was how much the categorical variables 'clarity' and 'color' contribute to the model. Adding clarity alone jumped adjusted R^2 from 0.9518 to 0.9846- the single biggest improvement of any variable added. This was somewhat surprising given that carat already explained ~93.6% of the variation on its own. It was also interesting that x, y, and z added virtually nothing to the model (only 0.0001 improvement) despite being strongly correlated with price in Part 1. This is because they measure the same underlying thing as carat- physical size- making them redundant once log(carat) is already in the model. This part reinforced that diamond pricing depends on carat, cut, color, and clarity, and that a well-transformed, carefully built model can explain nearly 98.5% of the variation in price.

Part III (Part II continued)

1. In class we saw different techniques and criterion to find best model. You can use any method and technique you prefer (e.g., backward elimination using AIC or stepwise regression using AIC or backward elimination using BIC criterion) to find the best model and document your observations.

The stepwise selection procedure was applied to the full dataset (n = 53,943) rather than the 1,500-observation sample, in order to obtain more stable AIC estimates. All conclusions carry over to the sample used in Parts 1 and 2. To determine the best regression model for predicting diamond price, a stepwise regression procedure using the AIC criterion was applied starting from the full model: price ~ carat + cut + color + clarity + depth + table + x + y + z

```r
library(MASS)
full_model <- lm(price ~ ., data = df)
step_model <- stepAIC(full_model, direction = "both")
```

```
Start:  AIC=758466.3
price ~ carat + cut + color + clarity + depth + table + x + y +
    z

          Df  Sum of Sq        RSS    AIC
- y         1 3.1529e+05 6.8858e+10 758465
```

```
<none>                  6.8858e+10 758466
- z         1 2.8614e+06 6.8861e+10 758467
- table     1 1.0556e+08 6.8963e+10 758547
- depth     1 2.5274e+08 6.9110e+10 758662
- cut       4 8.6368e+08 6.9721e+10 759131
- x         1 1.1998e+09 7.0057e+10 759396
- color     6 1.7081e+10 8.5939e+10 770408
- clarity   7 3.5703e+10 1.0456e+11 780986
- carat     1 6.8443e+10 1.3730e+11 795692

Step:  AIC=758464.6
price ~ carat + cut + color + clarity + depth + table + x + z

          Df  Sum of Sq        RSS    AIC
<none>                  6.8858e+10 758465
- z         1 2.6627e+06 6.8861e+10 758465
+ y         1 3.1529e+05 6.8858e+10 758466
- table     1 1.0581e+08 6.8964e+10 758545
- depth     1 2.5625e+08 6.9114e+10 758663
- cut       4 8.6420e+08 6.9722e+10 759129
- x         1 1.5415e+09 7.0399e+10 759657
- color     6 1.7082e+10 8.5940e+10 770406
- clarity   7 3.5708e+10 1.0457e+11 780987
- carat     1 6.8523e+10 1.3738e+11 795721
```

```
1 summary(step_model)
```

```
Call:
lm(formula = price ~ carat + cut + color + clarity + depth +
    table + x + z, data = df)

Residuals:
     Min       1Q   Median       3Q      Max
-21379.1   -592.4   -183.6    376.4  10694.2

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   2197.807    407.141   5.398 6.76e-08 ***
carat        11257.901     48.601 231.641  < 2e-16 ***
cutGood        580.366     33.571  17.288  < 2e-16 ***
cutIdeal       833.415     33.394  24.957  < 2e-16 ***
```

```
cutPremium     762.304     32.225  23.656  < 2e-16 ***
cutVery Good   727.419     32.213  22.581  < 2e-16 ***
colorE        -209.219     17.892 -11.694  < 2e-16 ***
colorF        -272.860     18.092 -15.082  < 2e-16 ***
colorG        -482.042     17.716 -27.210  < 2e-16 ***
colorH        -980.251     18.835 -52.043  < 2e-16 ***
colorI       -1466.263     21.162 -69.288  < 2e-16 ***
colorJ       -2369.415     26.130 -90.678  < 2e-16 ***
clarityIF     5345.419     51.019 104.774  < 2e-16 ***
claritySI1    3665.714     43.629  84.019  < 2e-16 ***
claritySI2    2702.880     43.814  61.690  < 2e-16 ***
clarityVS1    4578.719     44.541 102.799  < 2e-16 ***
clarityVS2    4267.433     43.849  97.321  < 2e-16 ***
clarityVVS1   5008.041     47.155 106.203  < 2e-16 ***
clarityVVS2   4951.118     45.850 107.986  < 2e-16 ***
depth          -63.986      4.517 -14.165  < 2e-16 ***
table          -26.496      2.911  -9.103  < 2e-16 ***
x            -1000.416     28.794 -34.743  < 2e-16 ***
z              -47.930     33.193  -1.444    0.149
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1130 on 53920 degrees of freedom
Multiple R-squared:  0.9198,    Adjusted R-squared:  0.9198
F-statistic: 2.811e+04 on 22 and 53920 DF,  p-value: < 2.2e-16
```

The stepwise procedure compares models by removing or adding predictors and selecting the model with the lowest AIC value. From the output, removing y produced the smallest AIC value (758464.6), which is lower than the AIC of the full model (758466.3). Therefore, the variable y was removed from the model. The resulting reduced model is:

price ~ carat + cut + color + clarity + depth + table + x + z

This model was selected as the best model based on the AIC criterion because it provides a slightly better balance between model fit and model complexity. Examining the regression output for the selected model shows that most predictors are statistically significant ($p < 0.001$), indicating strong relationships with diamond price. The only variable that appears insignificant is z ($p = 0.149$), suggesting that this predictor contributes less to explaining variation in price compared with the other variables. The model explains a very large proportion of the variability in diamond prices, with:

$R^2 = 0.9198$ Adjusted $R^2 = 0.9198$

This indicates that approximately 92% of the variation in diamond prices is explained by the predictors included in the model, demonstrating that the selected regression model provides a

very strong fit to the data. It is worth noting that the variable z is retained in the selected model despite a p-value of 0.149, which falls above the conventional 0.05 significance threshold. The stepwise AIC procedure retained z because AIC optimizes the bias-variance tradeoff rather than strict significance testing — removing z did not improve AIC. However, z should be interpreted cautiously, as its individual contribution is weak once the other predictors are included.

2. Detect multicollinearity among the variables using the variance inflation factor (VIF)

```r
library(car)
vif(step_model)
```

```
              GVIF Df GVIF^(1/(2*Df))
carat    22.416227  1        4.734578
cut       1.933407  4        1.085901
color     1.178308  6        1.013767
clarity   1.347182  7        1.021515
depth     1.768902  1        1.330001
table     1.787048  1        1.336805
x        44.067405  1        6.638328
z        23.175906  1        4.814136
```

To detect multicollinearity among the predictors, the Variance Inflation Factor (VIF) was calculated for the variables in the selected model. The adjusted VIF values $GVIF^{1/2Df}$ were examined since several predictors are categorical with multiple degrees of freedom.

Most predictors (cut, color, clarity, depth, and table) have adjusted VIF values close to 1, indicating little to no multicollinearity. The variables carat (4.73), x (6.64), and z (4.81) show relatively higher values, suggesting moderate multicollinearity. This is expected because the physical dimensions of a diamond (carat, x, y, z) are naturally related to each other.

However, since the adjusted VIF values remain below the commonly used threshold of 10, there is no evidence of severe multicollinearity, and the predictors can remain in the model.

3. Give CIs for a mean predicted value and the PIs of a future predicted value for at least one combination of X's (from your final linear model)

```r
new_diamond <- data.frame(
  carat = 1,
  cut = "Ideal",
  color = "E",
  clarity = "VS1",
```

```
  depth = 61,
  table = 57,
  x = 6.5,
  z = 4
)

predict(step_model, newdata = new_diamond, interval = "confidence")
```

```
      fit      lwr      upr
1 6550.77 6512.871 6588.668
```

```
predict(step_model, newdata = new_diamond, interval = "prediction")
```

```
      fit      lwr      upr
1 6550.77 4335.516 8766.024
```

The confidence interval provides the estimated range for the mean diamond price for diamonds with these characteristics. The prediction interval provides the range where the price of a single future diamond with these characteristics is expected to fall.

As expected, the prediction interval is wider than the confidence interval because it includes both the uncertainty in estimating the mean and the variability of individual diamond prices.

4. Summarize your report (for the final deliverable).

This project analyzed the Diamonds dataset to understand how different characteristics influence diamond prices. A random sample of 1,500 observations was used throughout Parts 1 and 2, with the final model selection in Part 3 conducted on the full dataset for stability.

In Part 2, a simple linear regression of price on carat revealed a strong positive relationship, but diagnostic plots showed clear violations of the linearity, normality, and constant variance assumptions. Applying a log transformation to both price and carat resolved all three violations, and the transformed model explained approximately 93.5% of the variation in log(price). Adding depth, table, cut, color, and clarity one at a time each improved adjusted R^2, with clarity providing the largest single gain.

The final model from Part 2 was log(price) ~ log(carat) + depth + table + cut + color + clarity, with an adjusted R^2 of 0.9833. In Part 3, stepwise regression using the AIC criterion on the full dataset selected the model price ~ carat + cut + color + clarity + depth + table + x + z after removing y. This model achieves an R^2 of approximately 0.92.

Multicollinearity was assessed using the VIF; most predictors showed values close to 1, and although carat, x, and z had moderately higher values due to their shared relationship with

diamond size, all adjusted VIF values remained below 10, indicating no severe multicollinearity.

For a diamond with 1 carat, Ideal cut, E color, VS1 clarity, depth of 61, table of 57, x of 6.5, and z of 4, the predicted price was \$6,550.77. The 95% confidence interval for the mean price of diamonds with these characteristics was [\$6,512.87, \$6,588.67], and the 95% prediction interval for an individual diamond was [\$4,335.52, \$8,766.02]. The prediction interval is wider because it accounts for both model uncertainty and individual variability. Overall, carat weight, cut, color, and clarity are the dominant drivers of diamond price, and the regression models built in this project provide strong and reliable predictions.